

PHASE 1

1.WHAT IS SOC?

SOC Workflow

L1 Analyst-

L2 Analyst-

L3 Analyst-

2.SIEM

3.LOG

4.TP , FP & BTP

5.MTTD & MTTR

6.INCIDENT RESPONSE

Six Stages

7.SHIFT HANDOVER

8.MITRE ATT&CK FRAMEWORK

9.CYBER KILL CHAIN

10.LOG SOURCES

WINDOWS EVENT LOGS

LINUX LOGS

FIREWALL LOGS

CLOUD AUDIT LOGS

11.ALERT TRIAGE

12.Severity Levels

1.WHAT IS SOC?

SOC-Security Operation Centre , its a **team** responsible for monitor alerts , detecting threats , investigating incidents , respond to attack , protect org.

SOC Workflow

```
Log Generated
↓
SIEM Receives Log
↓
Rule Triggers Alert
↓
L1 Analyst Investigates
```

↓
TP / FP / BTP
↓
Escalate or Close

Escalate when:

- Confirmed malicious activity
- Successful unauthorized access
- Critical asset affected
- Data exfiltration suspected
- L2/IR action required

L1 Analyst-

- Monitor alerts
- Initial investigation
- Classify alerts
- Document findings
- Escalate necessary

L2 Analyst-

- Deep investigation
- malware analysis
- incident handling
- root cause analysis

L3 Analyst-

- Threat hunting
- Detection engineering
- Advanced investigation

- SOC improvements

2.SIEM

SIEM-Security information and event management

- Collect logs
- Correlates events
- Detects threats
- Generate alerts

Eg: Wazuh , Splunk , QRadar , Sentinel

3.LOG

A log is a record of activity

Eg: User logged in , Files deleted , Password changed , Program executed

4.TP , FP & BTP

True Positive	False Positive	Benign True Positive
Alert is real.	Alert says attack. Investigation says it's harmless.	Activity is real. But authorized.
Eg: Brute-force attack detected	Eg: Security scanner triggers alert	Eg: IT team runs vulnerability scan

5.MTTD & MTTR

Mean Time To Detect	Mean Time To Respond
Time taken to discover a threat.	Time taken to respond after detection.
Eg: Attack started: 10:00 AM Detected: 10:15 AM	Eg: Detected: 10:15 AM Contained: 10:25 AM

6. INCIDENT RESPONSE

When a security incident occurs, we follow a process.

Six Stages

1. Preparation
2. Detection & Analysis
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

7. SHIFT HANDOVER

SOCs run 24/7.

When your shift ends:

You must tell the next analyst:

- Open incidents
- Pending actions
- Critical alerts
- Environment status

8.MITRE ATT&CK FRAMEWORK

MITRE ATT&CK is a knowledge base of how attackers behave / tactics & techniques after compromising systems.

MITRE helps us describe:

- What they did
- How they did it
- Why they did it

Tactic = Attacker's goal

Technique = How attacker achieves that goal

9.CYBER KILL CHAIN

Another attack model used by SOC teams.

The 7 stages are:

Stage	Meaning
Reconnaissance	Gather information
Weaponization	Prepare attack
Delivery	Send payload
Exploitation	Trigger vulnerability
Installation	Install malware
Command & Control	Connect to attacker
Actions on Objectives	Steal data, move laterally

10.LOG SOURCES

WINDOWS EVENT LOGS

Common IDs:

Event ID	Meaning
4624	Successful Login
4625	Failed Login
4688	Process Created
4720	User Created
1102	Audit Logs Cleared

LINUX LOGS

Usually found in:

```
/var/log/auth.log  
/var/log/syslog
```

FIREWALL LOGS

Tell us:

- Source IP
- Destination IP
- Port
- Allowed or Blocked

CLOUD AUDIT LOGS

Examples:

- AWS CloudTrail
- Azure Activity Logs
- Google Cloud Audit Logs

Track:

- User actions

- Resource changes
- Login events

11.ALERT TRIAGE

Workflow:

```
Alert Received
  ↓
Validate Alert
  ↓
Gather Context
  ↓
Assess Impact
  ↓
Close or Escalate
```

12.Severity Levels

Severity	Meaning
Critical	Immediate response
High	Serious threat
Medium	Needs investigation
Low	Minor issue
Informational	For awareness